

PROJECT REPORT

IMPORT DATA USING TRANSFORM MAPS (Spreadsheet) - ServiceNow

Team ID : SWTID-2026-8221

Team Members

S.No	Name	Role
1	Pradeep N	Team Lead
2	Dushyanth O A	Member
3	Jagadeesh N	Member
4	Dharani T	Member
5	Praveen K	Member

B.Tech Artificial Intelligence & Data Science (Third Year)
Anna University Regional Campus, Coimbatore

INDEX

1. INTRODUCTION.....	3
1.1 Project Overview	3
1.2 Purpose	3
2. IDEATION PHASE	4
2.1 Problem Statement	4
2.2 Empathy Map Canvas.....	5
2.3 Brainstorming	5
3. REQUIREMENT ANALYSIS	7
3.1 Customer Journey map	7
3.2 Solution Requirement	7
3.3 Data Flow Diagram	9
3.4 Technology Stack	12
4. PROJECT DESIGN	15
4.1 Problem Solution Fit	15
4.2 Proposed Solution	16
4.3 Solution Architecture.....	18
5. PROJECT PLANNING & SCHEDULING	20
5.1 Project Planning	20
6. FUNCTIONAL AND PERFORMANCE TESTING	23
6.1 Performance Testing	23
7. RESULTS	25
7.1 Output Screenshots.....	25
8. ADVANTAGES & DISADVANTAGES.....	33
8.1 ADVANTAGES	33
8.2 DISADVANTAGES	33
9. CONCLUSION	33
10. FUTURE SCOPE	34
11. APPENDIX	35

1. INTRODUCTION

1.1 Project Overview

This micro project focuses on importing structured employee data from an external Excel spreadsheet into the ServiceNow platform. The project demonstrates the complete process of transferring bulk data into ServiceNow using Import Sets and Transform Maps. The employee data from the Excel spreadsheet is first loaded into an Import Set table, which acts as a temporary staging area for the imported information. A Transform Map is then configured to establish a relationship between the source fields and the corresponding fields in the target table. During the transformation process, the staged data is converted into usable records in ServiceNow. After the transformation is completed, the created records are verified to ensure that the employee information has been imported correctly. This project provides practical knowledge of data loading, field mapping, data transformation, record creation, and validation in ServiceNow.

1.2 Purpose

The main purpose of this project is to understand the basic concepts and procedures involved in data migration within the ServiceNow platform. It helps beginners learn how to import bulk employee data from an external Excel spreadsheet efficiently. The project provides an understanding of the role of Import Sets in temporarily storing imported data and the use of Transform Maps for mapping source fields to appropriate target fields. It also demonstrates how data transformation creates records in the target table and how the imported records can be verified for accuracy and data integrity. Overall, the project provides hands-on experience with the complete ServiceNow data import process, from data loading to transformation and final validation.

2. IDEATION PHASE

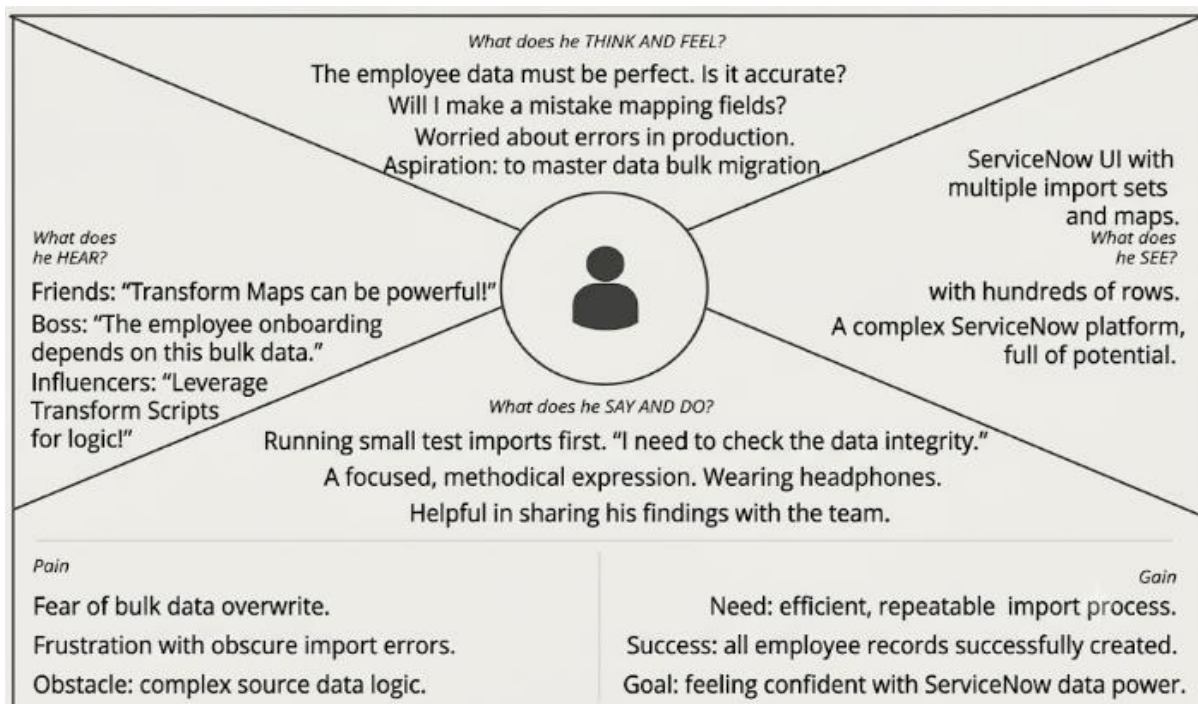
2.1 Problem Statement

Problem Statement:

This micro project focuses on importing structured data from an external spreadsheet into the ServiceNow platform using Import Sets and Transform Maps. The project simulates a real-world scenario where bulk employee data is received in Excel format and must be migrated into ServiceNow efficiently and accurately. The spreadsheet data is first loaded into an Import Set table, which serves as a staging area, and a Transform Map is then configured to map source fields to the appropriate target table fields. Upon executing the transformation, records are created in the target table and validated to ensure data integrity. This project helps beginners understand the end-to-end data import process in ServiceNow, including data loading, field mapping, transformation execution, and result verification.

I am	Describe customer with 3-4 key characteristics - <i>who are they?</i>	A ServiceNow platform beginner seeking a practical understanding of the complete end-to-end data import workflow (loading, mapping, transformation).
I'm trying to	List their outcome or "job" the care about - <i>what are they trying to achieve?</i>	To efficiently and accurately migrate bulk employee data from an external Excel spreadsheet into a production-ready ServiceNow target table.
but	Describe what problems or barriers stand in the way - <i>what bothers them most?</i>	The source data from the spreadsheet must first be loaded into a staging area (Import Set table) and cannot be directly ingested into the target table.
because	Enter the "root cause" of why the problem or barrier exists - <i>what needs to be solved?</i>	A Transform Map must be carefully configured to correctly align each source field from the staging table to its corresponding target field before the data can be transformed into records.
which makes me feel	Describe the emotions from the customer's point of view - <i>how does it impact them emotionally?</i>	A sense of relief and accomplishment, with newfound confidence in the platform's data capabilities, after successfully creating and validating data integrity for the new records.

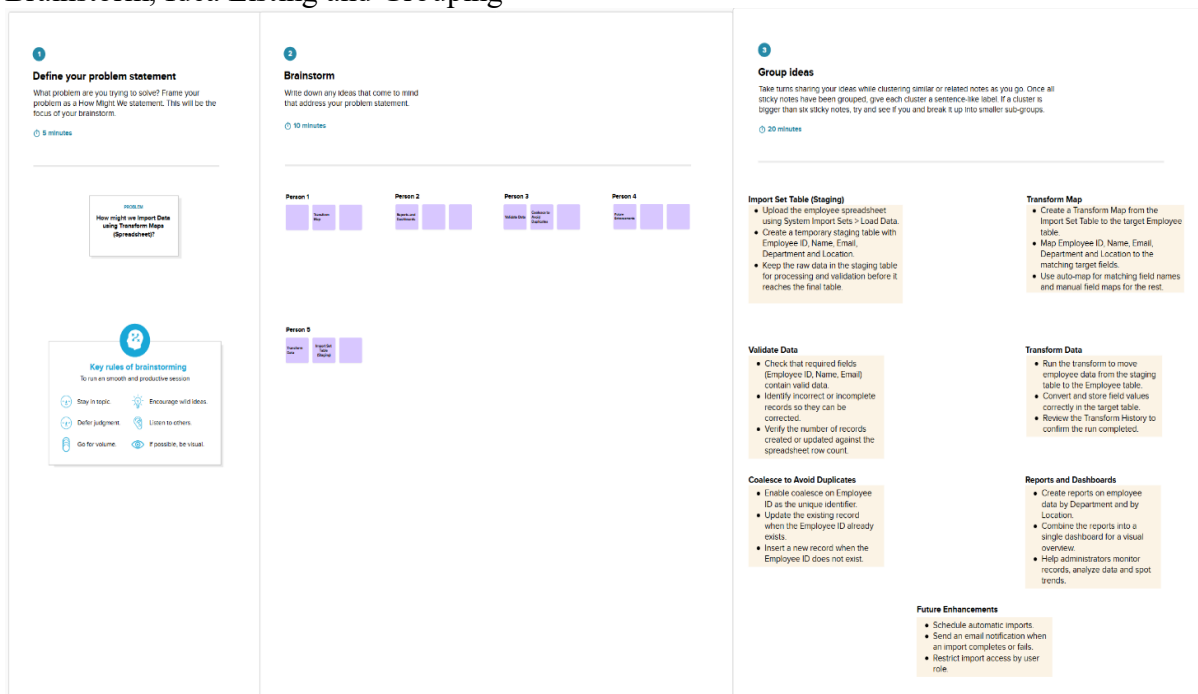
2.2 Empathy Map Canvas



2.3 Brainstorming

Step-1 & Step-2:

Team Gathering, Collaboration and Select the Problem Statement
Brainstorm, Idea Listing and Grouping



Step-3: Idea Prioritization

Idea	Impact	Feasibility	Priority
Import Set Table to stage the spreadsheet data	High	High	Core
Transform Map with field mapping	High	High	Core
Transform Data into the Employee table	High	High	Core
Validate Data (Employee ID, Name, Email)	High	High	Core
Coalesce on Employee ID to avoid duplicates	High	High	Core
Reports and Dashboard by Department and Location	Medium	High	Core
Transform script for extra validation	Medium	Medium	Secondary
Schedule automatic imports	Medium	Low	Future scope
Email notification on import completion	Low	Medium	Future scope
Role-based restriction on imports	Low	Medium	Future scope

Ideas selected for implementation (project milestones):

1. Create the Import Set Table and load the employee spreadsheet.
2. Configure the Transform Map to the target Employee table.
3. Run Transform Data to move the records.
4. Validate the imported data.
5. Enable Coalesce on Employee ID to avoid duplicate records.
6. Build Reports and a Dashboard for the imported employee data.

These six milestones cover the complete import lifecycle, from loading the spreadsheet to monitoring the results. Scheduling, notifications and role-based access are recorded as future scope.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map



3.2 Solution Requirement

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-01	Spreadsheet Import	The system shall allow the administrator to upload an Excel spreadsheet containing employee data.

FR-02	Data Validation	The administrator shall be able to verify required fields such as Employee ID, Name, Email, Department, and Location before importing.
FR-03	Import Set Processing	The system shall load the spreadsheet data into the Employee Import staging table.
FR-04	Transform Map Configuration	The administrator shall be able to create and configure a Transform Map between the Employee Import and Employee Test tables.
FR-05	Field Mapping	The system shall map Employee ID, Name, Email, Department, and Location from the source fields to the corresponding target fields.
FR-06	Duplicate Handling	The system shall use Employee ID as the Coalesce field to identify existing employee records.
FR-07	Record Insertion	The system shall create new employee records in the Employee Test table when no matching Employee ID exists.
FR-08	Record Update	The system shall update existing employee records when a matching Employee ID is found.
FR-09	Transformation Execution	The administrator shall be able to execute the Transform Map and process the imported employee data.
FR-10	Transformation Results	The system shall record inserts, updates, skipped records, and errors in Transform History .
FR-11	Result Verification	The administrator shall be able to verify the transformed records in the Employee Test table and validate the imported data.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Category	Requirement
NFR-01	Usability	The import process should be simple and easy for administrators to perform.
NFR-02	Security	Only authorized users should be able to upload, transform, and manage employee data.
NFR-03	Reliability	The system should process valid employee data consistently without data loss.
NFR-04	Performance	Spreadsheet data should be processed efficiently within a reasonable time.

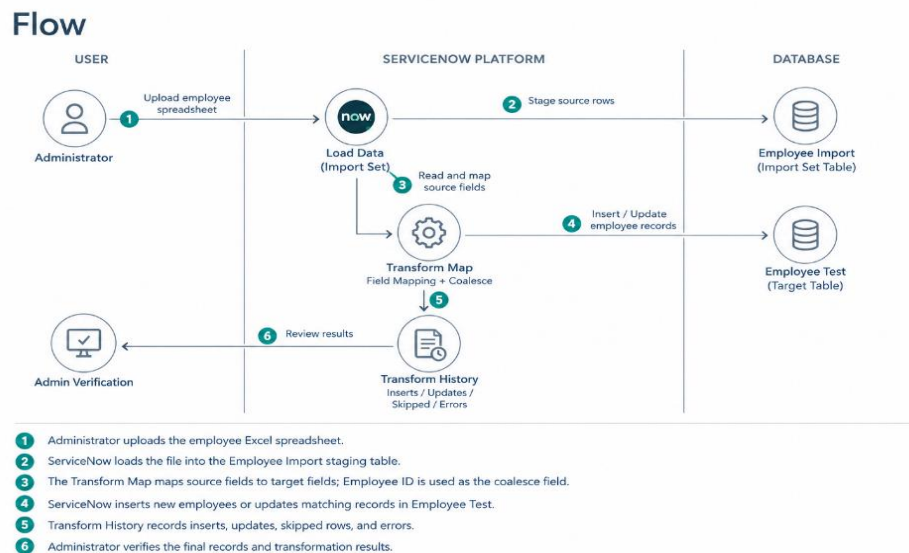
NFR-05	Availability	The import and transformation features should be available whenever the ServiceNow platform is accessible.
NFR-06	Scalability	The solution should support increasing numbers of employee records.
NFR-07	Data Integrity	Employee data should remain accurate and consistent during import and transformation.

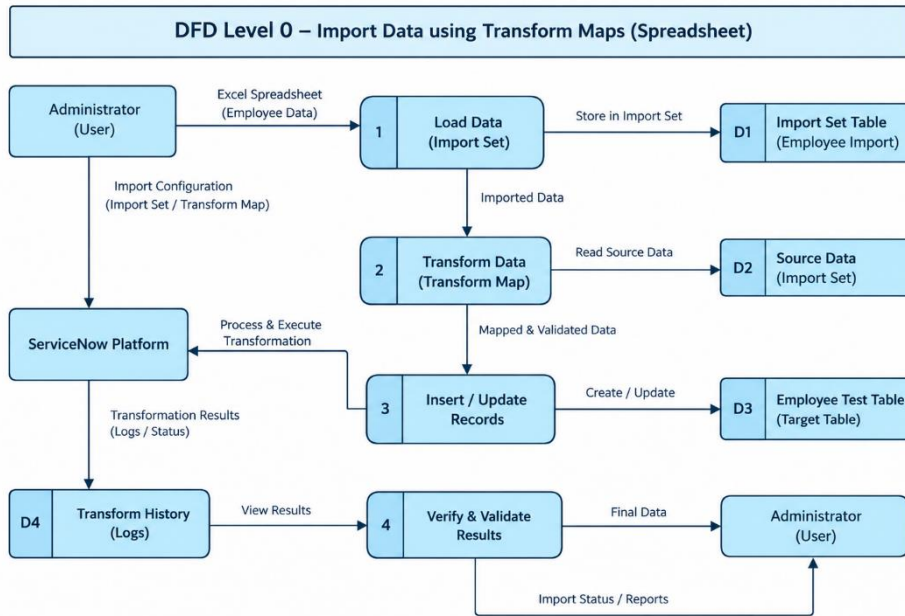
3.3 Data Flow Diagram

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Data flow and DFD Level 0 of the proposed solution:





User Stories

User Type	Functional Requirement (Epic)	User Story No	User Story / Task	Acceptance Criteria	Priority	Release
Administrator	Spreadsheet Import	USN-1	As an administrator, I can upload an Excel spreadsheet containing employee data into ServiceNow.	I can upload the .xlsx file and the employee data is loaded successfully.	High	Sprint-1
		USN-2	As an administrator, I can verify the employee spreadsheet before importing it.	The spreadsheet contains Employee ID, Name, Email, Department, and Location fields in the required format.	High	Sprint-1
	Import Set Processing	USN-3	As an admin, I can load the spreadsheet data into the Employee Import staging table.	The imported rows are displayed in the Employee Import table without errors.	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story No	User Story / Task	Acceptance Criteria	Priority	Release
	Transform Map	USN-4	As an admin, I can create a Transform Map between the Employee Import source table and Employee Test target table.	The Transform Map shows Employee Import as the source and Employee Test as the target.	High	Sprint-1
	Field Mapping	USN-5	As an admin, I can map the employee source fields to the corresponding target fields.	Employee ID, Name, Email, Department, and Location are mapped correctly.	High	Sprint-1
	Duplicate Handling	USN-6	As an admin, I can use Employee ID as the coalesce field to identify existing employee records.	Matching Employee IDs update existing records instead of creating duplicate records.	High	Sprint-2
	Record Insertion	USN-7	As an administrator, I can insert new employee records into the Employee Test table through the Transform Map.	New Employee IDs create new records in the Employee Test table.	High	Sprint-2
	Record Update	USN-8	As an administrator, I can update existing employee records using the Transform Map.	Existing records with matching Employee IDs are updated with the latest spreadsheet values.	High	Sprint-2

User Type	Functional Requirement (Epic)	User Story No	User Story / Task	Acceptance Criteria	Priority	Release
	Transformation Execution	USN-9	As an administrator, I can execute the Transform Map to process the imported employee data.	The transformation completes successfully and displays the processing status.	High	Sprint-2
	Result Verification	USN-10	As an administrator, I can verify the transformation results using Transform History.	Transform History displays the number of inserts, updates, skipped records, and errors.	High	Sprint-2
	Error Handling	USN-11	As an administrator, I can identify records that were skipped or failed during transformation.	Skipped and error records can be identified from the transformation results.	Medium	Sprint-2
	Data Validation	USN-12	As an administrator, I can verify that the transformed employee data is correctly stored in the Employee Test table.	Target records match the corresponding valid source data.	High	Sprint-2

3.4 Technology Stack

Technical Architecture:

The project uses the **ServiceNow cloud platform** to import employee data from an Excel spreadsheet. The data is first loaded into an **Import Set staging table**, then processed through a **Transform Map**. Field mappings and the Employee ID coalesce field are used to create or update records in the **Employee Test** target table. Transformation results are recorded in **Transform History**.

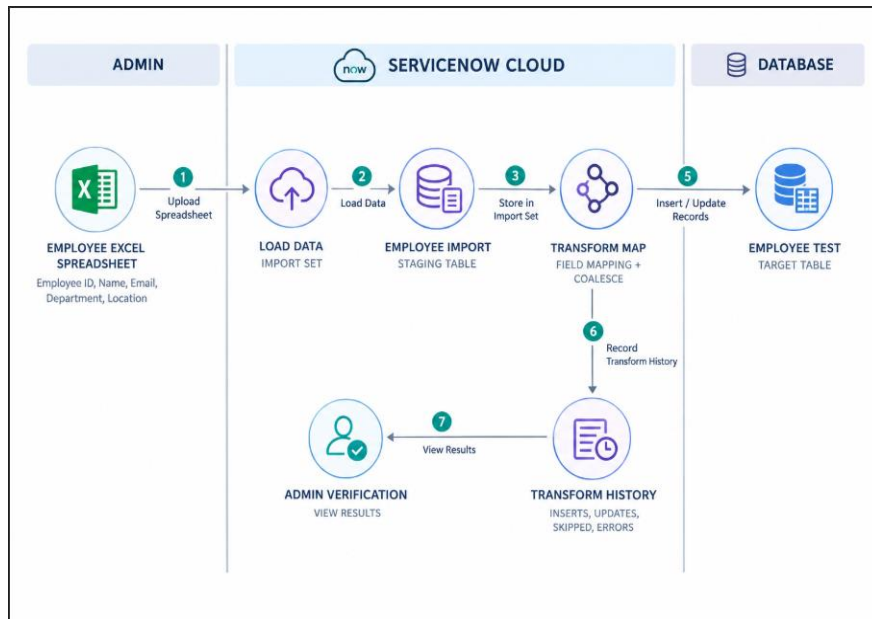


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Interface used by the administrator to upload and manage employee data.	ServiceNow Web UI
2	Application Logic-1	Loads employee data from the spreadsheet into the staging table.	ServiceNow Import Sets
3	Application Logic-2	Maps source fields to corresponding target fields.	ServiceNow Transform Map
4	Application Logic-3	Identifies existing records and performs insert or update operations.	Coalesce – Employee ID
5	Database	Stores the final transformed employee records.	ServiceNow Platform Database
6	Cloud Database	Provides database storage for the ServiceNow application.	ServiceNow Cloud Database
7	File Storage	Stores the employee spreadsheet used for importing data.	Excel .xlsx File
8	Machine Learning Model	No machine learning model is used in this project.	Not Applicable
9	Infrastructure (Server / Cloud)	Hosts the application, data processing, and database services.	ServiceNow Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No external open-source frameworks are used in the project.	ServiceNow Platform
2	Security Implementations	Access to employee data and import functions is controlled through authorized users, roles, access controls.	ServiceNow Roles & ACLs
3	Scalable Architecture	Import Sets and Transform Maps support processing of increasing volumes of employee records.	ServiceNow Import Sets & Transform Maps
4	Availability	The application is available through the ServiceNow cloud platform when the platform is accessible.	ServiceNow Cloud Platform
5	Performance	Staging employee data before transformation helps process spreadsheet records efficiently.	Import Sets & Transform Maps

References:

1. ServiceNow Developers – Importing Data.

https://developer.servicenow.com/print_page.do?category=learningmodule&identifier=app_store_learnv2_importingdata_xanadu_importing_data_objectives%2Cnew_to_servicenow&module=learning+plan&release=xanadu

2. ServiceNow Developers – Create a Transform Map.

https://developer.servicenow.com/print_page.do?category=learningplans&identifier=servicenow_application_developer&module=learning+plan&release=yokohama

3. ServiceNow Developer – Import Series Part 1: Getting Started with Import Sets.

<https://developer.servicenow.com/blog.do?p=%2Fpost%2Fimport-series-part-1---getting-started-with-import-sets%2F>

4. ServiceNow Developer – Import Series Part 2: Anatomy of an Import Set.

<https://developer.servicenow.com/blog.do?p=%2Fpost%2Fimport-series-part-2-anatomy-of-an-import-set%2F>

4. PROJECT DESIGN

4.1 Problem Solution Fit

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- Solve complex problems in a way that fits the state of your customers.
- Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- Sharpen your communication and marketing strategy with the right triggers and messaging.
- Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- Understand the existing situation in order to improve it for your target group.

1. CUSTOMER SEGMENT(S) [CS] - IT administrators and ServiceNow platform admins - HR / IT operations staff who onboard employees - Beginner ServiceNow developers handling data migration	6. CUSTOMER CONSTRAINTS [CC] - Data arrives only as Excel files - Limited time and no coding skills for beginners - Must not create duplicate or corrupt records - Access restricted to admin roles	5. AVAILABLE SOLUTIONS [AS] - Manual entry record by record – slow and error-prone - Spreadsheet copy-paste into list view – limited and risky - Custom scripts / integrations – need coding effort and maintenance
2. JOBS-TO-BE-DONE / PROBLEMS [J&P] - Move bulk employee data from Excel into ServiceNow quickly - Map spreadsheet columns to the correct table fields - Confirm that all records were imported correctly	9. PROBLEM ROOT CAUSE [RC] - Source data lives outside ServiceNow in spreadsheets - Column names and formats differ from table fields - No staging step to review data before it goes live - No simple method to verify data integrity after loading	7. BEHAVIOUR [BE] - Retype or copy-paste rows one at a time - Re-check records by eye after entry - Ask experienced colleagues for help - Search documentation and tutorials for import steps
3. TRIGGERS [TR] - A new batch of employees joins and needs accounts - HR sends an updated spreadsheet - Seeing a colleague lose hours on manual entry 4. EMOTIONS: BEFORE / AFTER [EM] - Before: stressed, worried about mistakes and duplicates - After: confident and in control – data is accurate and traceable	10. YOUR SOLUTION [SL] - Load the spreadsheet into an Import Set (staging) table - Use a Transform Map to map source fields to target fields with a coalesce field - Run the transform to insert/update records - Verify results using Transform History, import log and list-view checks - Result: fast, accurate, repeatable bulk import with no custom code	8. CHANNELS of BEHAVIOUR [CH] 8.1 ONLINE: ServiceNow Import Sets module, Data Source and Transform Map forms, ServiceNow docs and community, video tutorials 8.2 OFFLINE: Email with spreadsheet from HR, team discussions, training sessions and review meetings

References:

- <https://www.idealhackers.network/problem-solution-fit-canvas/>
- <https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>

4.2 Proposed Solution

S.No.	Parameter	Description
1.	Problem Statement	Employee data is received from HR/other systems as Excel spreadsheets with hundreds of rows. Entering this data manually into ServiceNow is slow, repetitive and error-prone (typos, duplicate records, missing or mis-typed fields), and there is no easy way to confirm that every row reached the platform correctly. A reliable, repeatable and verifiable way to bulk-load spreadsheet data into ServiceNow is needed.

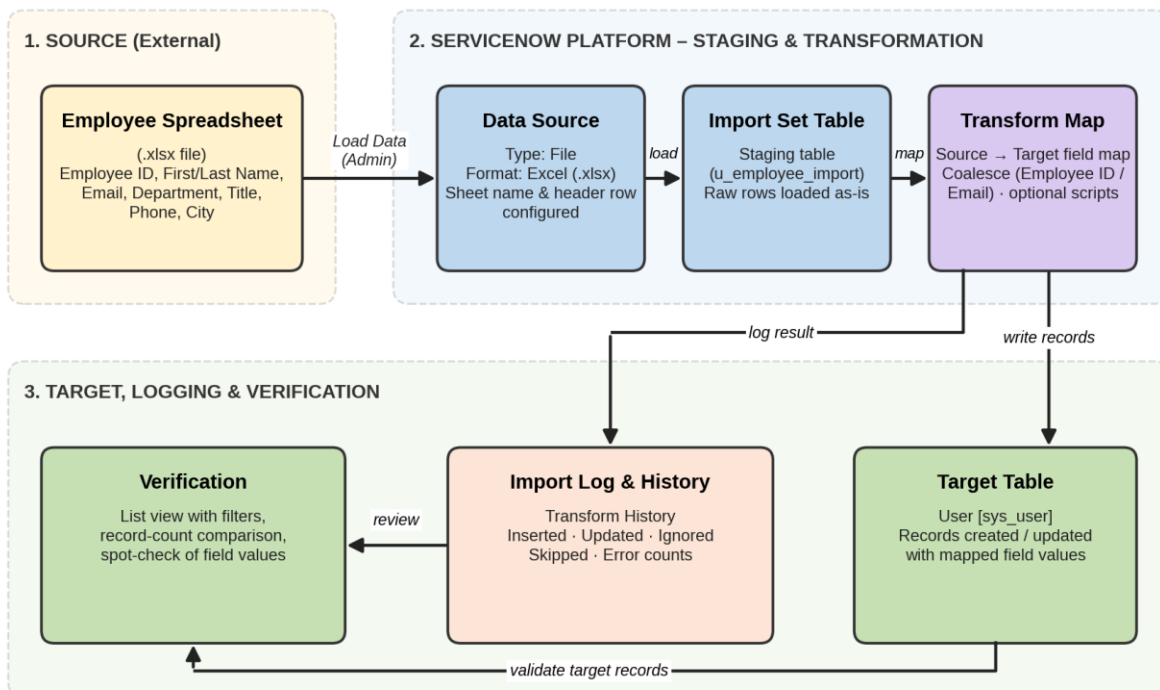
2.	Idea / Solution description	Use ServiceNow's native Import Set framework to load the spreadsheet into a staging (Import Set) table via a Data Source of type File / Excel. A Transform Map then maps each source column (Employee ID, First Name, Last Name, Email, Department, Title, Phone, City) to the matching field of the target table (User [sys_user]), with a coalesce field to avoid duplicates. Running the transform inserts or updates records in the target table, and the results are verified using Transform History, the import log and list-view checks (record counts and sample field values).
3.	Novelty / Uniqueness	The solution uses only out-of-the-box ServiceNow features – no external integration tool or custom code is needed. A staging table separates raw data from live data, so bad rows can be inspected before they affect production tables. Coalescing makes the import re-runnable (existing records are updated, not duplicated), and the built-in transform history gives a clear audit trail of inserted, updated, ignored and skipped rows.
4.	Social Impact / Customer Satisfaction	Saves time for administrators and HR/IT staff, reduces data-entry mistakes and gives users accurate, up-to-date records from day one. Faster onboarding of employees, consistent data for assignments, approvals and notifications, and higher trust in the platform's data improve overall customer (internal user) satisfaction.
5.	Business Model (Revenue Model)	Delivered as a value-adding service for ServiceNow implementation and managed-service engagements: (a) fixed-fee data migration / onboarding packages, (b) reusable import templates and transform maps licensed or bundled with implementation projects, and (c) ongoing support contracts for scheduled or periodic imports. The solution also reduces the internal labour cost of manual data entry.
6.	Scalability of the Solution	Highly scalable: the same pattern works for any table (Incidents, CMDB CIs, Locations, Departments, Assets) and any file source (Excel, CSV, XML, JSON). Large files can be processed in batches, imports can be automated with scheduled data sources, and one transform map can be reused for every future file with the same layout. The approach grows from a few records to many thousands without changing the design.

4.3 Solution Architecture

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture – Spreadsheet Data Import using Import Sets & Transform Maps (ServiceNow)



Data flow description:

1. Source: The employee spreadsheet (.xlsx) containing Employee ID, First Name, Last Name, Email, Department, Title, Phone and City is prepared with a clean header row.
2. Data Source & Load: The administrator creates a Data Source (Type: File, Format: Excel), attaches the spreadsheet and uses Load Data to create the Import Set table.
3. Staging: The rows are stored unchanged in the Import Set table (u_employee_import), which acts as a staging area so the raw data can be checked before it touches live tables.
4. Transformation: A Transform Map links the Import Set table to the target table (User [sys_user]). Each source field is mapped to a target field, and a coalesce field (Employee ID / Email) prevents duplicates.

5. Execution: Running the transform inserts new records and updates existing ones in the target table.
6. Logging: Transform History and the import log record the number of inserted, updated, ignored, skipped and errored rows.
7. Verification: The target table list view is filtered and compared with the spreadsheet (record count and sample field values) to confirm data integrity.

Components and technologies:

Component	Purpose
Excel spreadsheet (.xlsx)	External source of bulk employee data
Data Source (sys_data_source)	Defines file type, format, sheet and header row
Import Set table (u_employee_import)	Staging table that holds raw imported rows
Transform Map (sys_transform_map)	Field mappings, coalesce rule and optional scripts
Target table – User (sys_user)	Final destination of validated employee records
Transform History / Import Log	Audit trail and error analysis of each run

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Requirement (Epic)	User Story No	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Milestone 1: Creation Of Spreadsheet And Table	USN-1	As an admin, I can create a spreadsheet containing employee details (Employee ID, Name, Email, Department, Location) that will be imported into ServiceNow.	2	High	Jagadeesh
Sprint-1		USN-2	As an admin, I can create the target Employee table with fields for Employee ID, Name, Email, Department and Location.	3	High	Dushyanth
Sprint-1	Milestone 2: Creation Of Import Set Table And Transform Map	USN-3	As an admin, I can create an Import Set table by loading the spreadsheet, so the data is staged before it is transformed.	3	High	Dharani
Sprint-1		USN-4	As an admin, I can create a Transform Map that maps the Import Set fields to the matching fields of the Employee table.	5	High	Praveen
Sprint-2	Milestone 3: Transform Data & Validate And Enable Coalesce To Avoid Duplicate Records	USN-5	As an admin, I can run the transform and validate that employee records are created correctly in the target table.	5	High	Pradeep
Sprint-2		USN-6	As an admin, I can enable coalesce on Employee ID so an existing employee is updated instead of a duplicate being created.	3	High	Pradeep
Sprint-2		USN-7	As an admin, I can insert new data in Excel format and import it, so new employees are added and existing employees are updated.	2	Medium	Dushyanth
Sprint-2	Milestone 4: Creation Of Reports & Dashboards	USN-8	As an admin, I can create Report 1 (employees by department) from the Employee table.	3	Medium	Jagadeesh
Sprint-2		USN-9	As an admin, I can add the reports to a dashboard to get a single view of the imported employee data.	2	Medium	Dharani

Project Tracker, Velocity & Burndown Chart

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	13	6 Days	5 Oct 2026	10 Oct 2026		
Sprint-2	15	6 Days	12 Oct 2026	17 Oct 2026		

Velocity:

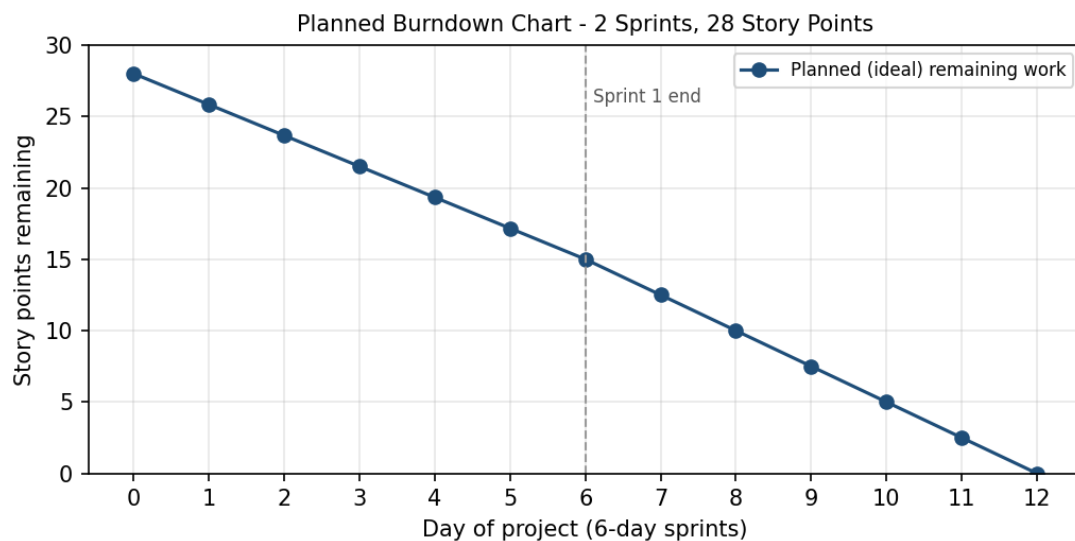
Velocity is the number of story points a team completes in one sprint. Dividing it by the sprint duration gives the average velocity (AV) per iteration unit, i.e. story points per day.

$$AV = \frac{\text{sprint duration}}{\text{velocity}} = \frac{20}{10} = 2$$

Velocity (this project): Sprint duration = 6 days; Sprint-1 = 13 story points and Sprint-2 = 15 story points. Team velocity = $(13 + 15) / 2 = 14$ story points per sprint. Average velocity (AV) = $14 / 6 \approx 2.33$ story points per day.

Burndown Chart: A burndown chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum, but it can be applied to any project with measurable progress over time.

Planned burndown for this project (28 story points over two 6-day sprints). The actual remaining points can be plotted on the same chart as the work progresses.



References:

<https://www.visual-paradigm.com/scrum/scrum-burndown-chart/>

<https://www.atlassian.com/agile/project-management>

<https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software>

<https://www.atlassian.com/agile/tutorials/epics>

<https://www.atlassian.com/agile/tutorials/sprints>

<https://www.atlassian.com/agile/project-management/estimation>

<https://www.atlassian.com/agile/tutorials/burndown-charts>

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

User Acceptance Testing (UAT)

Project Description: Employee data received in an Excel spreadsheet is imported into ServiceNow using an Import Set table (staging) and a Transform Map into a target Employee table. Employee ID is the coalesce field, so existing employees are updated and new employees are inserted. Reports and a dashboard give an overview of the imported data.

Testing Scope: Creation of the employee spreadsheet and Employee table; Import Set table and Transform Map; transform, validation and coalesce on Employee ID; inserting new data in Excel format; Report 1 and dashboard. User stories USN-1 to USN-9 (Milestones 1 to 4 of the project plan).

Testing Environment: ServiceNow instance - [https://\[your-instance\].service-now.com](https://[your-instance].service-now.com) (admin user).

Test Cases

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-001	Load a valid spreadsheet into an Import Set table	1. Go to System Import Sets > Load Data 2. Select Employee_Data.xlsx, sheet 1, header row 1 3. Submit	Import Set table is created and 15 rows are loaded	Import Set table created; Processed: 15, inserts 15, errors 0 (State: Complete, Success).	Pass
TC-002	Transform loaded data into the Employee table	1. Open the loaded import set 2. Click Transform and choose the Employee Transform Map 3. Open the Employee table list	15 employee records are created; Transform History shows 15 inserts and 0 errors	Transformation completed successfully; Employee Test list shows 15 of 15 records.	Pass
TC-003	Verify field values after transform	1. Open employee SB-0001 2. Compare Name, Email, Department and Location with the spreadsheet	All values match the spreadsheet exactly	SB-0001 shows Ratan Kumar, test1@gmail.com, ServiceNow, Hyderabad - matches the spreadsheet row.	Pass
TC-004	Coalesce updates an existing employee	1. Load Employee_Data_Update.xlsx 2. Transform 3. Open employee SB-0010	SB-0010 shows the updated values from the spreadsheet (Ravi Teja, test18@gmail.com); no second SB-0010 record exists	Import set rows 0 and 1 show state Updated; SB-0010 shows Ravi Teja, test18@gmail.com; only one SB-0010 record exists.	Pass

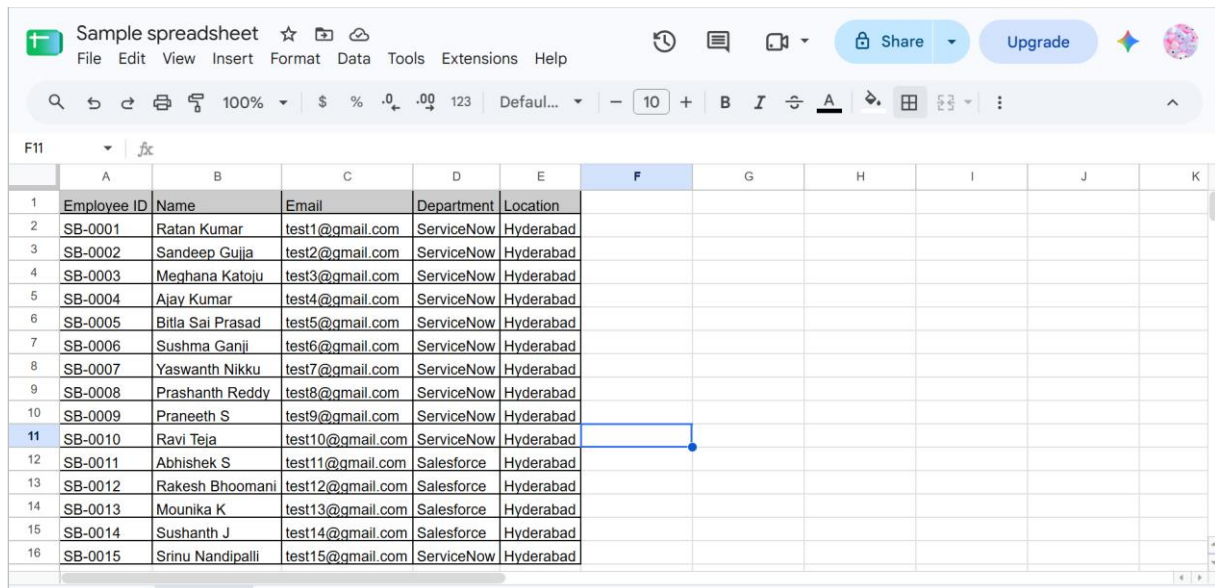
Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-005	No duplicate records on re-import (negative)	1. Transform the same import set a second time 2. Check the Employee list for repeated Employee IDs	No duplicate Employee IDs; record count does not increase	Transform History for ISET0010014: Total 4, Inserts 0, Updates 0, Ignored 4, Errors 0; no duplicate IDs.	Pass
TC-006	New employees are inserted	1. After TC-004, filter the Employee list for SB-0016 and SB-0017	Two new records exist; total employee records = 17	Import set rows 2 and 3 show state Inserted; SB-0016 and SB-0017 exist; list shows 17 of 17 records.	Pass
TC-007	Row with missing required data (negative)	1. Prepare a file with one row whose Email is blank 2. Load and transform it	Row is ignored and logged as an error; no incomplete record is created	Not yet executed.	Pending
TC-008	Report 1 shows correct data	1. Open Report 1 (Employees by Department) 2. Compare the counts with the Employee list grouped by Department	Counts per department match the Employee table	Report shows ServiceNow = 13 and Salesforce = 4 (total 17), matching the Employee list.	Pass
TC-009	Dashboard displays the reports	1. Open the dashboard 2. Check that Report 1 is displayed	Dashboard loads and shows the report with current data	Dashboard loads and shows Employee List, Employees by Location and Employees by Department (Report 1).	Pass

Bug Tracking

Bug ID	Bug Description	Steps to reproduce	Severity	Status	Additional feedback
BG-001					

7. RESULTS

7.1 Output Screenshots



	A	B	C	D	E	F	G	H	I	J	K
1	Employee ID	Name	Email	Department	Location						
2	SB-0001	Ratan Kumar	test1@gmail.com	ServiceNow	Hyderabad						
3	SB-0002	Sandeep Gujja	test2@gmail.com	ServiceNow	Hyderabad						
4	SB-0003	Meghana Katoju	test3@gmail.com	ServiceNow	Hyderabad						
5	SB-0004	Ajay Kumar	test4@gmail.com	ServiceNow	Hyderabad						
6	SB-0005	Bitla Sai Prasad	test5@gmail.com	ServiceNow	Hyderabad						
7	SB-0006	Sushma Ganji	test6@gmail.com	ServiceNow	Hyderabad						
8	SB-0007	Yaswanth Nikku	test7@gmail.com	ServiceNow	Hyderabad						
9	SB-0008	Prashanth Reddy	test8@gmail.com	ServiceNow	Hyderabad						
10	SB-0009	Praneeth S	test9@gmail.com	ServiceNow	Hyderabad						
11	SB-0010	Ravi Teja	test10@gmail.com	ServiceNow	Hyderabad						
12	SB-0011	Abhishek S	test11@gmail.com	Salesforce	Hyderabad						
13	SB-0012	Rakesh Bhoomani	test12@gmail.com	Salesforce	Hyderabad						
14	SB-0013	Mounika K	test13@gmail.com	Salesforce	Hyderabad						
15	SB-0014	Sushanth J	test14@gmail.com	Salesforce	Hyderabad						
16	SB-0015	Srinu Nandipalli	test15@gmail.com	ServiceNow	Hyderabad						

Figure 7.1: Employee spreadsheet (Employee_Data.xlsx) used as the import source

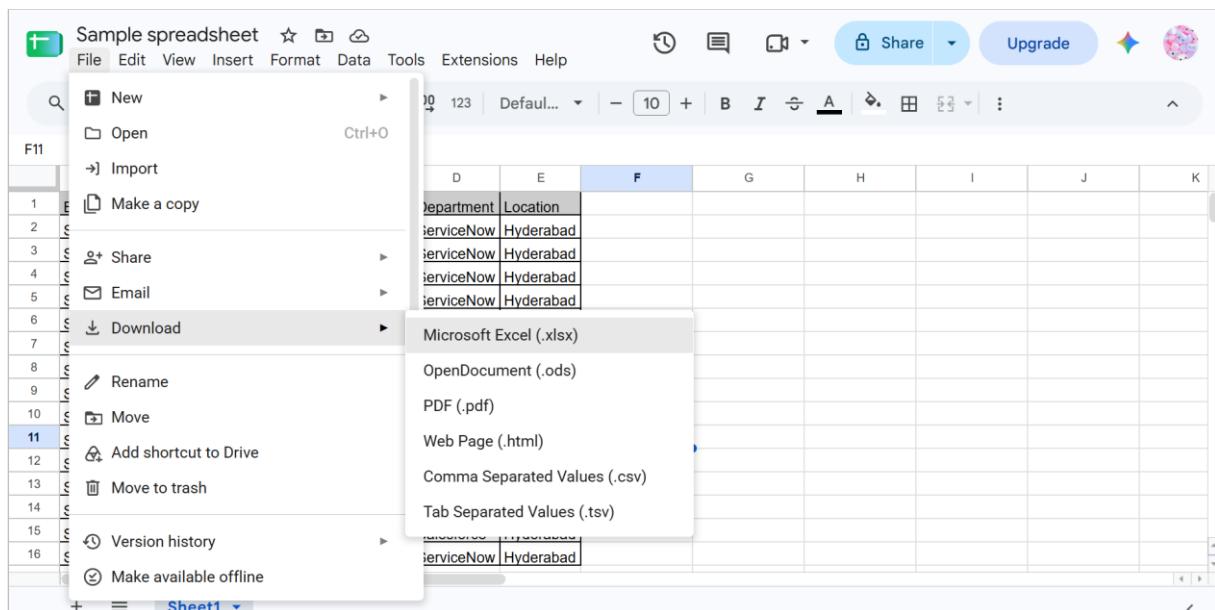


Figure 7.1 (continued): Spreadsheet downloaded in Microsoft Excel (.xlsx) format

The screenshot shows a web form titled "Employee Test" with a subtitle "New record". The form contains five input fields: "Employee ID", "Employee Name", "Email", "Department", and "Location". A "Submit" button is located at the bottom left of the form area. The "Employee ID" field is currently active, indicated by a green border.

Figure 7.2: Employee target table and its fields

The "Personalize List Columns" dialog box is shown. It has two main sections: "Available" and "Selected". The "Available" section lists "Created", "Created by", "Tags", "Updated", "Updated by", and "Updates". The "Selected" section lists "Employee ID", "Employee Name", "Email", "Department", and "Location". There are arrows between the sections to move items. At the bottom, there are checkboxes for "Wrap column text" (checked), "Compact rows" (unchecked), "Active row highlighting" (unchecked), "Modern cell coloring" (checked), "Enable list edit" (checked), and "Double click to edit" (checked). There are also buttons for "Reset to column defaults", "Cancel", and "OK".

Figure 7.2 (continued): Employee Test table columns (Employee ID, Employee Name, Email, Department, Location)

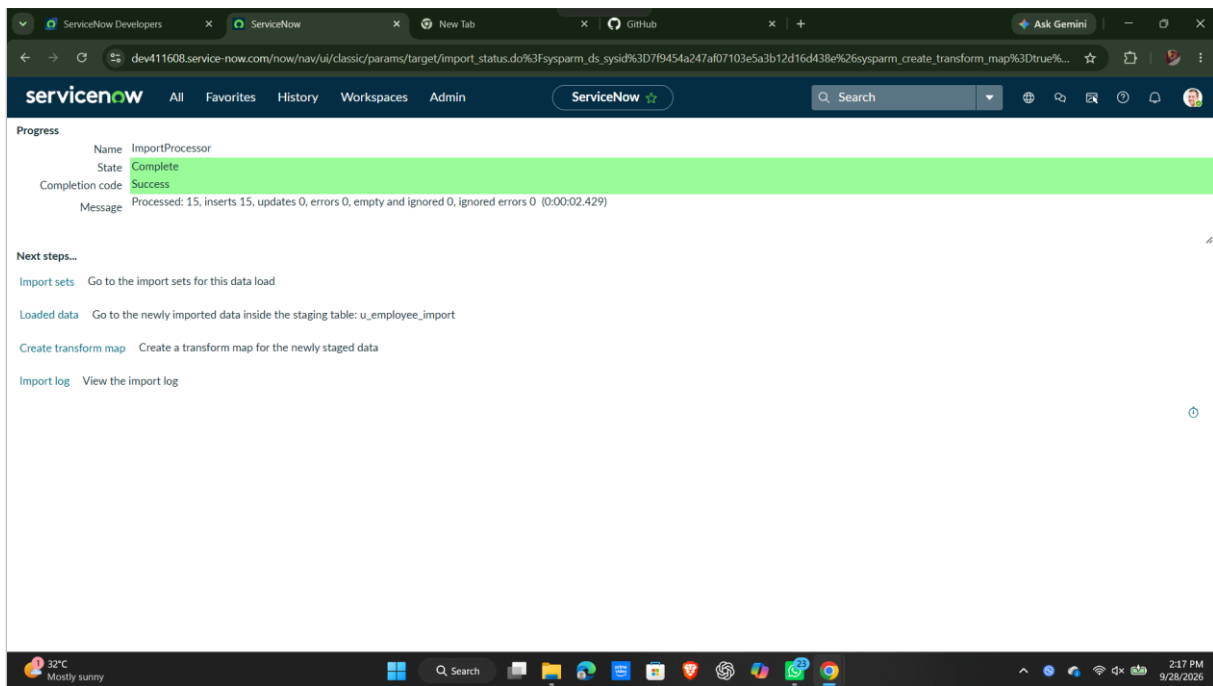


Figure 7.3: Import Set table loaded with the spreadsheet rows (Load Data)

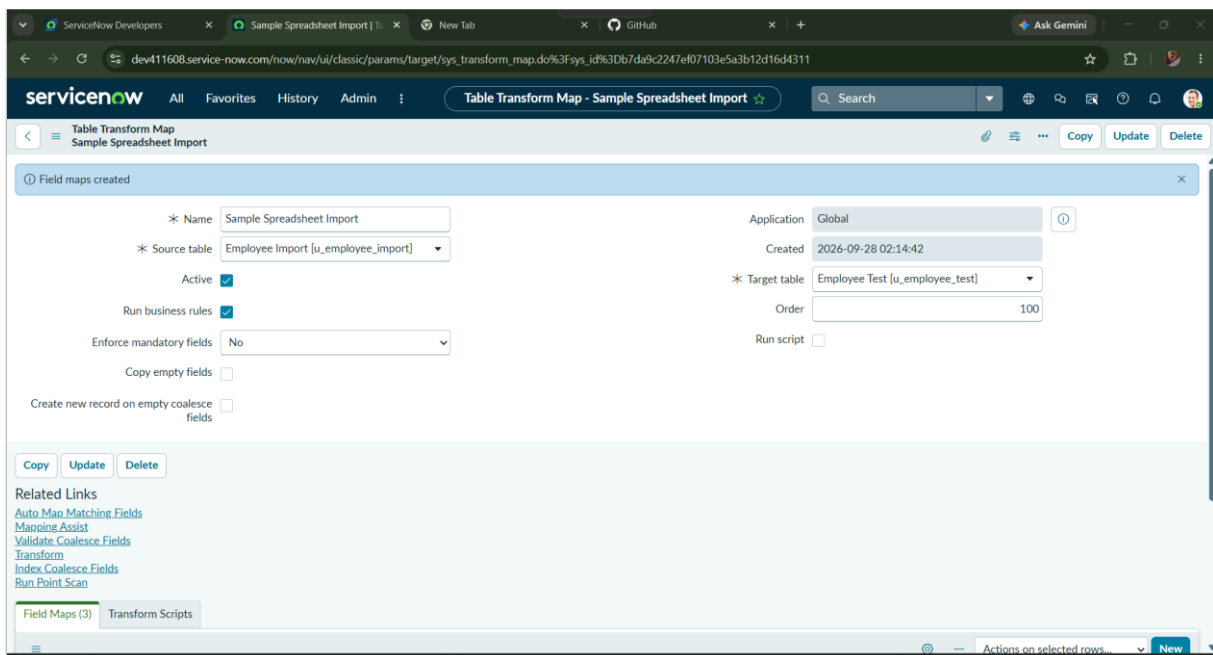


Figure 7.4: Transform Map with source-to-target field mapping and Employee ID coalesce

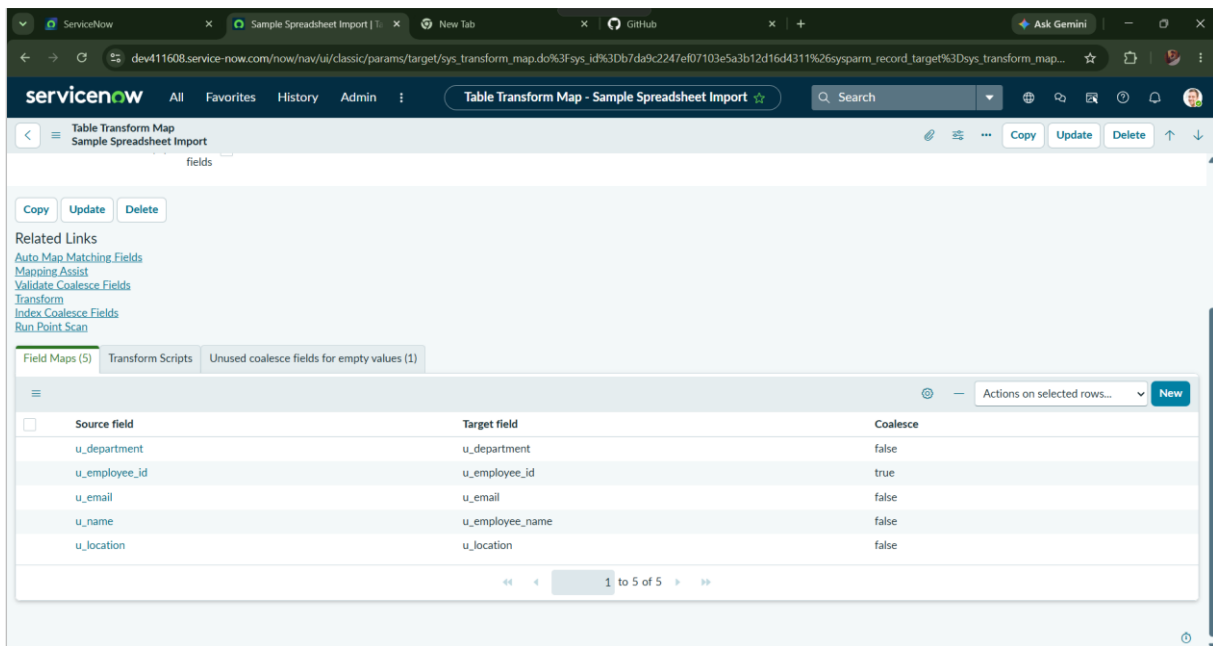


Figure 7.4 (continued): Field maps with Employee ID set as the coalesce field

Name	Source table	Target table	Run business rules	Order	Active	Updated
Auth0 Example	label [oidc_auth0_import]	User [sys_user]	true	100	true	2018-02-02 14:04:26
Azure AD Example	label [oidc_azure_ad_import]	User [sys_user]	true	100	true	2018-02-02 14:18:38
Sample Spreadsheet Import	Employee Import [u_employee_import]	Employee Test [u_employee_test]	true	100	true	2026-09-28 04:43:37
Notification	Notification [imp_notification]	Incident [incident]	true	100	true	2008-12-28 22:08:39
User	User [imp_user]	User [sys_user]	true	100	true	2015-07-07 07:32:42
Location	Location [imp_location]	Location [cmn_location]	true	100	true	2015-07-07 07:30:52
Computer	Computer [imp_computer]	Computer [cmdb_ci_computer]	true	100	true	2015-07-07 07:29:19
ADFS Example	label [oidc_adfs_import]	User [sys_user]	true	100	true	2018-02-02 10:58:24
Google Example	label [oidc_google_import]	User [sys_user]	true	100	true	2018-02-06 13:42:44
Employee Job Profile Transform Map	Employee Job Profile Staging [sn_employee_position_stg]	Employee Job Profile [sn_employee_position]	true	100	true	2023-02-21 21:00:47
Target Import	Target Import Set [imp_sn_gf_goal_target]	Target [sn_gf_goal_target]	true	100	true	2022-06-29 08:41:47
Okta Example	label [oidc_okta_import]	User [sys_user]	true	100	true	2018-02-02 14:21:44

Figure 7.4 (continued): Transform Maps list showing Sample Spreadsheet Import (Employee Import to Employee Test)

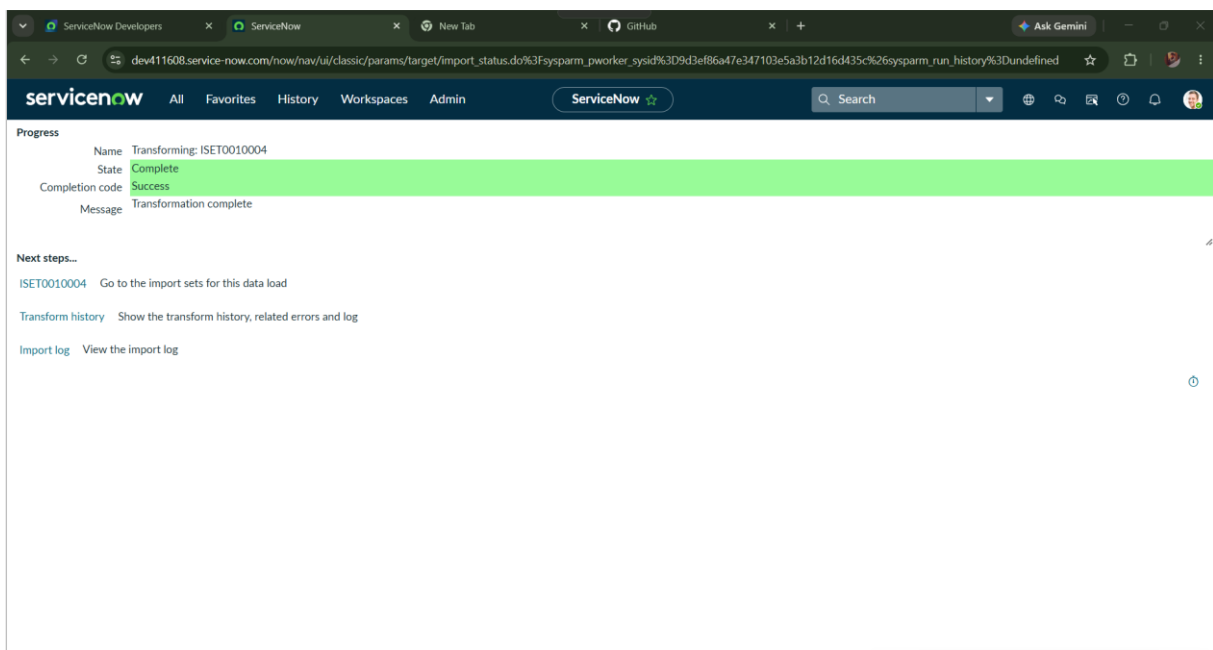


Figure 7.5: Transform Data result and Transform History (inserts, updates, errors)

The screenshot displays the 'Employee Tests' table in ServiceNow. The table contains 15 rows of data, including Employee ID, Employee Name, Email, Department, and Location. The table is filtered by 'Department' and 'Search'. The 'Employee ID' column shows values like SB-0014, SB-0011, SB-0012, etc. The 'Employee Name' column shows names like Sushanth J, Abhishek S, Rakesh Bhoomani, etc. The 'Email' column shows email addresses like test14@gmail.com, test11@gmail.com, test12@gmail.com, etc. The 'Department' column shows 'Salesforce' and 'ServiceNow'. The 'Location' column shows 'Hyderabad'. The table is sorted by 'Employee ID' and the 'Department' column is expanded.

Employee ID	Employee Name	Email	Department	Location
SB-0014	Sushanth J	test14@gmail.com	Salesforce	Hyderabad
SB-0011	Abhishek S	test11@gmail.com	Salesforce	Hyderabad
SB-0012	Rakesh Bhoomani	test12@gmail.com	Salesforce	Hyderabad
SB-0013	Mounika K	test13@gmail.com	Salesforce	Hyderabad
SB-0002	Sandeep Guja	test2@gmail.com	ServiceNow	Hyderabad
SB-0006	Sushma Ganji	test6@gmail.com	ServiceNow	Hyderabad
SB-0010	Ravi Teja	test10@gmail.com	ServiceNow	Hyderabad
SB-0015	Srinu Nandipalli	test15@gmail.com	ServiceNow	Hyderabad
SB-0003	Meghana Katoju	test3@gmail.com	ServiceNow	Hyderabad
SB-0007	Yaswanth Nikku	test7@gmail.com	ServiceNow	Hyderabad
SB-0008	Prashanth Reddy	test8@gmail.com	ServiceNow	Hyderabad
SB-0001	Ratan Kumar	test1@gmail.com	ServiceNow	Hyderabad
SB-0004	Ajay Kumar	test4@gmail.com	ServiceNow	Hyderabad
SB-0005	Bitla Sai Prasad	test5@gmail.com	ServiceNow	Hyderabad
SB-0009	Praneeth S	test9@gmail.com	ServiceNow	Hyderabad

Figure 7.6: Employee table list view after the transform

The screenshot displays the 'Transform Histories' table in ServiceNow. The table contains 1 row of data, showing the transform history for 'Employee Import [u_employee_import]'. The 'Started' column shows '2026-09-28 23:47:31', the 'State' is 'Complete', the 'Completed' column shows '2026-09-28 23:47:31', the 'Run time' is '0 Seconds', the 'Set' is 'ISET0010013', the 'Import set table' is 'Employee Import [u_employee_import]', the 'Total' is '4', the 'Inserts' is '2', the 'Updates' is '2', the 'Ignored' is '0', the 'Skipped' is '0', the 'Errors' is '0', and the 'Transform Map' is 'Sample Spreadsheet Import'.

Started	State	Completed	Run time	Set	Import set table	Total	Inserts	Updates	Ignored	Skipped	Errors	Transform Map
2026-09-28 23:47:31	Complete	2026-09-28 23:47:31	0 Seconds	ISET0010013	Employee Import [u_employee_import]	4	2	2	0	0	0	Sample Spreadsheet Import

Figure 7.7: Coalesce result: existing record updated and new employees inserted after the second import

The screenshot shows the 'Employee Imports' interface in ServiceNow. The table displays the following data:

Row	Set	State	Target table	Target record	Error
0	ISSET0010013	Updated	Employee Test [u_employee_test]	Employee Test: Created 2026-09-28 23:35:37	(empty)
1	ISSET0010013	Updated	Employee Test [u_employee_test]	Employee Test: Created 2026-09-28 23:35:37	(empty)
2	ISSET0010013	Inserted	Employee Test [u_employee_test]	Employee Test: Created 2026-09-28 23:47:31	(empty)
3	ISSET0010013	Inserted	Employee Test [u_employee_test]	Employee Test: Created 2026-09-28 23:47:31	(empty)

Related Links:
[Import Sets](#)
[Transform Maps](#)
[Transform History](#)
[Edit Web Service](#)

Figure 7.7 (continued): Import set rows showing Updated and Inserted records

The screenshot shows the 'Employee Tests' interface in ServiceNow. The table displays the following data:

Employee ID	Employee Name	Email	Department	Location
SB-0014		test14@gmail.com	Salesforce	Hyderabad
SB-0013		test13@gmail.com	Salesforce	Hyderabad
SB-0012		test12@gmail.com	Salesforce	Hyderabad
SB-0011		test11@gmail.com	Salesforce	Hyderabad
SB-0016	New Employee 1	test16@gmail.com	ServiceNow	Hyderabad
SB-0002		test2@gmail.com	ServiceNow	Hyderabad
SB-0010	Ravi Teja	test18@gmail.com	ServiceNow	Hyderabad
SB-0006		test6@gmail.com	ServiceNow	Hyderabad
SB-0001		test1@gmail.com	ServiceNow	Hyderabad
SB-0009		test9@gmail.com	ServiceNow	Hyderabad
SB-0005		test5@gmail.com	ServiceNow	Hyderabad
SB-0008		test8@gmail.com	ServiceNow	Hyderabad
SB-0004	Ajay	test4@gmail.com	ServiceNow	Hyderabad
SB-0017	New Employee 2	test17@gmail.com	ServiceNow	Hyderabad
SB-0007		test7@gmail.com	ServiceNow	Hyderabad
SB-0003		test3@gmail.com	ServiceNow	Hyderabad
SB-0015		test15@gmail.com	ServiceNow	Hyderabad

Figure 7.7 (continued): Employee Test list after the second import (17 records)

The screenshot shows the 'Transform Histories' interface in ServiceNow. The table displays the following data:

Started	State	Completed	Run time	Set	Import set table	Total	Inserts	Updates	Ignored	Skipped	Errors	Transform Map
2026-09-28 23:51:36	Complete	2026-09-28 23:51:37	1 Second	ISSET0010014	Employee Import [u_employee_import]	4	0	0	4	0	0	Sample Spreadsheet Import

Figure 7.7 (continued): Re-running the same import set: all 4 rows ignored, no duplicates created

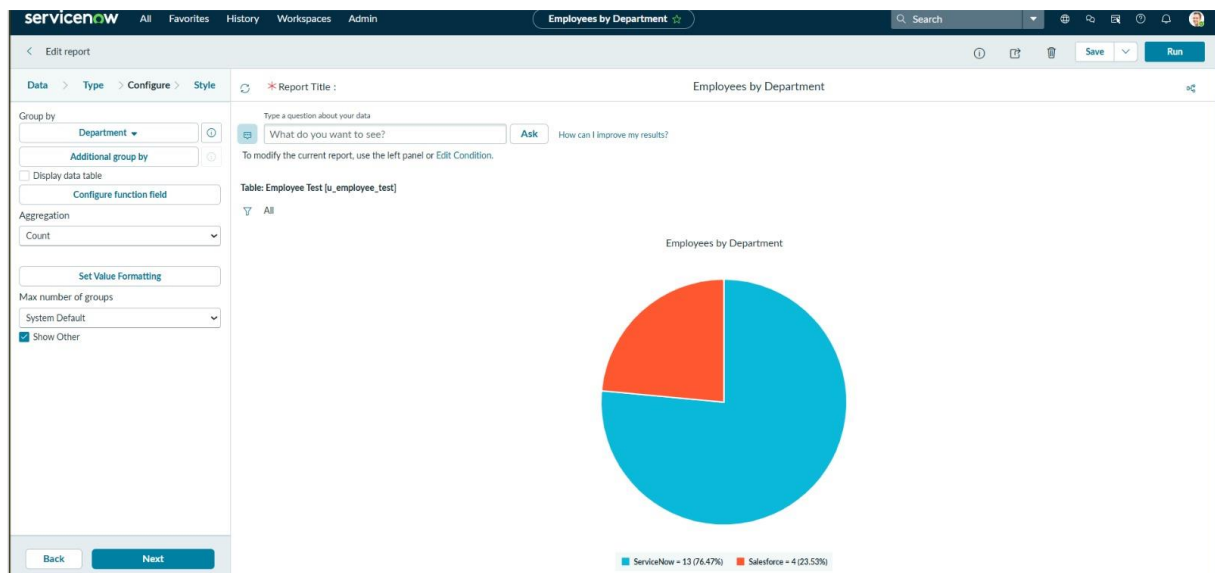


Figure 7.8: Report 1 - Employees by Department

Title	Table	Type	Field Name	Created by	Updated
Employees by Department	Employee Test [u_employee_test]	Pie	u_department	admin	2026-09-29 02:27:45
Employee List	Employee Test [u_employee_test]	List		admin	2026-09-29 02:30:17
Employees by Location	Employee Test [u_employee_test]	Horizontal bar	u_location	admin	2026-09-29 02:29:12

Figure 7.8 (continued): Reports list (Employees by Department, Employee List, Employees by Location)

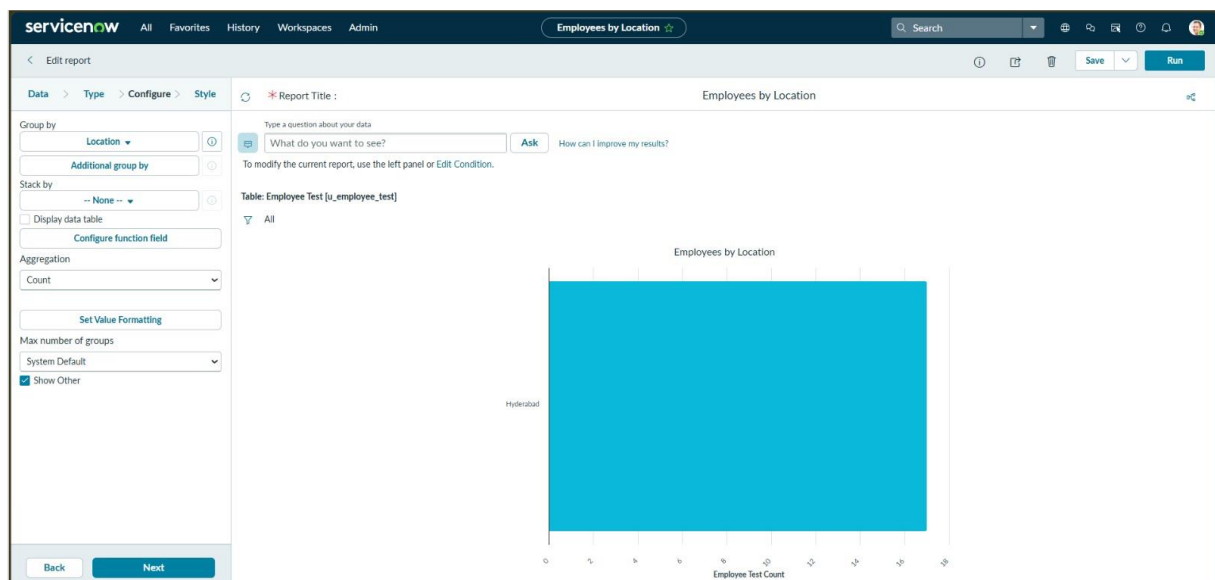


Figure 7.8 (continued): Additional report: Employees by Location

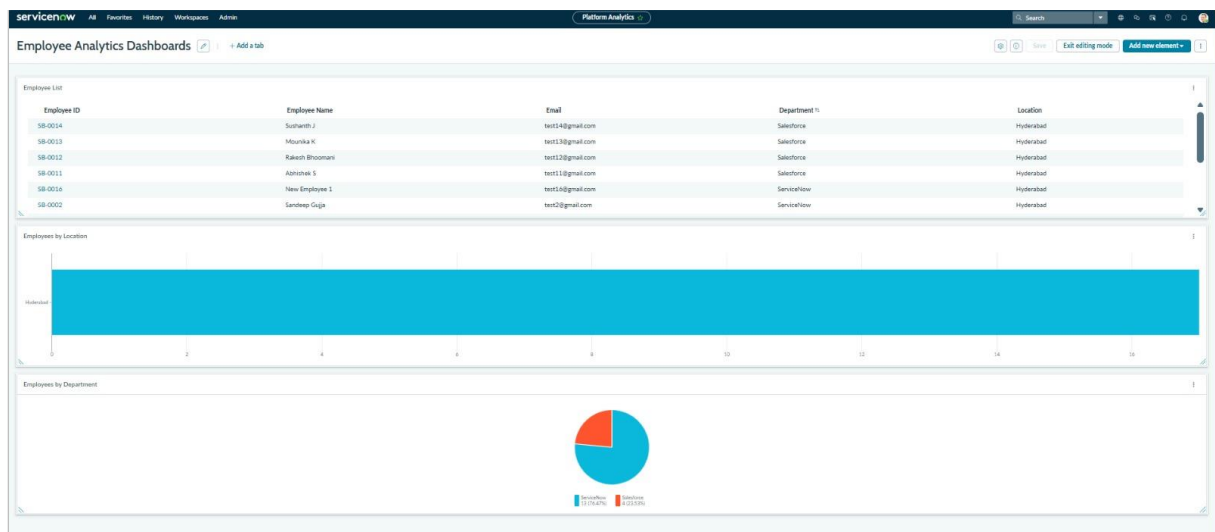


Figure 7.9: Dashboard showing the imported employee data

The screenshot shows the 'Dashboard record: Employee Analytics Dashboards' page. It includes the following sections:

- Form Fields:**
 - Name:** Employee Analytics Dashboards
 - Group:** (Empty field with a search icon)
 - Order:** (Empty field)
 - Restrict to roles:** (Icon to edit roles)
 - Active:** ☐
 - Owner:** System Administrator
- Buttons:** Update, Delete
- Related Links:** View Dashboard
- Breakdown Source:** Dashboard Tabs (1)
- Table:**
 - Header: Dashboard = Employee Analytics Dashboards
 - Columns: Tab, Order
 - Row 1: New Tab 1, 1
- Footer:** 1 to 1 of 1

Figure 7.9 (continued): Dashboard record: Employee Analytics Dashboards

8. ADVANTAGES & DISADVANTAGES

8.1 ADVANTAGES

- **Efficient Bulk Data Import** – Large amounts of employee data can be imported into ServiceNow at once instead of entering records manually.
- **Reduced Manual Work** – The automated transformation process reduces repetitive data-entry tasks and saves time.
- **Accurate Field Mapping** – Transform Maps allow source spreadsheet fields to be mapped to the appropriate target fields in ServiceNow.
- **Data Validation** – Imported records can be reviewed and validated after transformation to identify incorrect or missing information.
- **Beginner-Friendly Learning** – The project provides practical experience with important ServiceNow concepts such as Import Sets, Transform Maps, field mapping, and data migration.

8.2 DISADVANTAGES

- **Data Quality Dependency** – Incorrect, incomplete, or inconsistent data in the Excel spreadsheet can lead to errors during the import process.
- **Initial Configuration Required** – Import Sets and Transform Maps must be configured correctly before the data can be transformed.
- **Mapping Errors** – Incorrect field mapping may place employee information in the wrong ServiceNow fields.
- **Limited for Complex Data** – Very complex data relationships or advanced migration requirements may require additional scripting and configuration.
- **Duplicate Records Risk** – If appropriate matching or coalesce conditions are not configured, the transformation may create duplicate records.

9. CONCLUSION

This project successfully demonstrates the end-to-end process of importing employee data from an external Excel spreadsheet into ServiceNow using Import Sets and Transform Maps. The project shows how external data can be loaded into a staging table, mapped to the required target fields, transformed, and converted into ServiceNow records. It also highlights the importance of proper field mapping, data validation, and verification to maintain data accuracy and integrity. Through this project, beginners can gain practical knowledge of ServiceNow data migration and understand how bulk data can be handled efficiently. The concepts learned in this project can also be applied to real-world ServiceNow implementation and administration tasks.

10. FUTURE SCOPE

The project can be further enhanced to support more advanced and automated data migration requirements in ServiceNow. Future development can include importing data from multiple sources such as CSV files, databases, APIs, and external enterprise applications. Automated data validation and error-handling mechanisms can be added to identify missing, invalid, or duplicate records before transformation. Scheduled imports can also be implemented to regularly update employee information without manual intervention. The project can be extended by using Transform Scripts for advanced data conversion and business rules for additional validation. Integration with external systems through REST or SOAP APIs can further improve data synchronization. Additionally, dashboards and reports can be developed to monitor import results, errors, and data quality, making the solution more suitable for larger real-world ServiceNow environments.

11. APPENDIX

Dataset Link: [Dataset link](#)

GitHub Link: [Import-Data-using-Transform-Maps-Spreadsheet](#)

Project Demo Link: